



TEXAS TECH UNIVERSITY

Department of Physics & Astronomy

ASTR 3300-002/5300-005: Probability, Machine Learning and AI in Physical Sciences

SCHEDULE

9:30 – 10:50 am MW at ASB 210

INSTRUCTOR

Dr. Nihan Pol | Sci 017 | npol@ttu.edu

Office hours: 11:00 am - 12:30 pm MW or by appointment via email.

DESCRIPTION

The purpose of this course is to equip you with the tools necessary to understand and extract meaning from real world datasets. Real datasets are usually messy; they are not uniformly sampled in time, have gaps in them, the calibration uncertainties are not always fully specified, and they might be susceptible to unknown noise processes. However, several techniques already exist to deal with such data and we will cover these in this course. As such, this course will be focused on getting practical experience with implementing these methods in Python, and we will not dwell too much on the philosophy of inference. While we will draw a lot of examples from astronomy in this course, the methods are much more general and applicable to data from other fields, such as particle and condensed matter physics, engineering, and finance.

LEARNING OUTCOMES AND TOPICS

- The role of probability in inference.
- Frequentist and Bayesian inference.
- Bayesian parameter estimation and model selection using MCMC.
- Exploratory data analysis and visualization.
- Regression analysis.
- Time-series analysis.
- Machine learning and neural networks.

COURSE PREREQUISITES

While there are no firm pre-requisites for taking this course, please understand that this is an upper-level course and the work expectations during the course will be at that level. We list a few example pre-requisites courses that will give you a sense of what knowledge you

are expected to possess, though courses with equivalent coverage will also be considered. In addition to these pre-requisites, it will be desirable, though not required, for you to have some experience with programming in Python and basic Git and GitHub knowledge. These pre-requisites may be waived on a case-by-case basis; please reach out to Dr. Nihan Pol if you would like to discuss the waivers.

Reference pre-requisites: ASTR 2401, PHYS 2302, PHYS 3301, and PHYS 4325 or MATH 3350, or MATH 3354.

RESOURCES

Recommended:

Textbook: “Statistics, Data Mining, & Machine Learning in Astronomy” – Z. Ivezić, A.J. Connolly, J. T. VanderPlas, & A. Gray., Updated Edition (course textbook)

- ISBN: 9780691198309.
- You can purchase this directly from the Follett bookstore or any other retailer.

Other useful texts:

- “Bayesian Logical Data Analysis for the Physical Sciences” – P.C. Gregory
- “Modern Statistical Methods for Astronomy” – E.D. Feigelson & G. J. Babu
- “Bayesian Data Analysis” – A. Gelman, J. Carlin, H. Stern, D. Bunson, A. Vehtari, D. Rubin
 - Free: <http://www.stat.columbia.edu/~gelman/book>
- “Practical Statistics for Astronomers”— J. V. Wall & C. R. Jenkins
- “Python Data Science Handbook”— J. T. VanderPlas
 - Free: <https://github.com/jakevdp/PythonDataScienceHandbook>
- “Information theory, inference, and learning algorithms”— D. MacKay
 - Free: <https://www.inference.org.uk/mackay/itila/book.html>
- “Data analysis recipes: Fitting a model to data”— D. Hogg, J. Bovy, D. Lang
 - Arxiv: <https://arxiv.org/abs/1008.4686>
- “Data analysis recipes: Probability calculus for inference”— D. Hogg
 - Arxiv: <https://arxiv.org/abs/1205.4446>

WEB TOOLS

Canvas. The course web page is (or will soon be) on the Canvas system. Course announcements will be posted on this site. Canvas will also be used for electronic communications, and to post other relevant course material (such as lecture notes, grades, homework assignments etc.). Please check Canvas and your Texas Tech email at least once a day to ensure you are not missing important course information.

ASSESSMENT AND GRADING

The final grade is based on class participation (30%), homework (30%), and a final exam (40%). The details for these are given in their respective sections below. There is no extra credit or other means of altering your score.

My grading philosophy is weighted in favor of you showing that you understand the problem(s). Be as explicit as you can in all your work, and show all steps in the calculation to receive full credit. If you know your final answer is wrong or incomplete, say so! This tells me that you understand the material.

You are strongly encouraged to confer with your classmates on homework assignments, but I expect the work you submit to be your own. I will easily be able to use git tools to check whether code has been copied. Cases of cheating will lead to an automatic zero points on the assignment.

All assignments (but not their grades) will be available to the rest of the class in the GitHub repository. My hope is that you will learn collaboratively from each other's approaches. Science is a collaborative endeavor: This statement is a fair warning that your work will be freely available to all other students in the class. Sharing knowledge and expertise in this way is intended in the spirit of scientific collaboration among your classmates. But to reiterate, I will easily be able to use git tools to check whether code has been copied

The thresholds for letter grades are 85% for A, 75% for B, 65% for C, and 60% for D.

Late submission of assignments or final exam

Barring special arrangements made in advance of the due date, late submissions of portions of the final exam will not be accepted for credit. Barring prior arrangements, late submissions for lecture notebooks and homework assignments will be subject to the following deductions: 1 day late -25%, 2 days late -50%, 3 days late -75%, 4+ days late will not be accepted for credit.

LECTURES and CLASS PARTICIPATION

The first part of lectures will be a lecture format, with me explaining the material. The second part of the lecture will be an individual and collaborative problem-solving session using Jupyter notebooks and Python running on your personal laptop computers (or any equivalent machine). If you do not have a personal machine, please get in touch with me as soon as possible.

Participation credit will be assigned by submitting your completed copy of the lecture Jupyter notebook with the required tasks highlighted therein. The deadline for submitting these completed lecture Jupyter notebooks will be 11:59 pm CT the day of each lecture. Credit will be given for making a reasonable attempt at completing all the tasks in the Jupyter notebook.

HOMEWORK

Homework assignments will be available to access on Fridays, and will be due by 11:59 pm CT the Friday of the following week. We will be using GitHub to submit homework assignments. A demonstration for how to do this will be given in class during the first week of classes. There will be 9 homework assignments.

To get the highest grade on the homework, all solutions must contain suitable explanation or commentary to describe the methods that were used to find the solution. Your code will be judged based on (i) how well it is commented, (ii) how well it is structured, (iii) how well it is made compact and optimized, (iv) its speed, and (v) its efficacy in delivering the correct answer.

A lack of explanation or a severe deficit in one of the criteria above will lead to a point deduction.

Final Exam

This course will culminate in a final exam featuring a few problems that will employ the techniques learned throughout the semester applied to real world problems. The final will be conducted in two parts:

- (i) a take-home portion of the exam, intended to allow for time and thought in finding the solution to the problems posed in the exam. The expected output from this portion will be a script (Python+any other required language) that provides the answers to the problem;
- (ii) an oral presentation, where in a 1-on-1 setting with me, you will explain the solution that you have derived and answer any follow-up or clarification questions that I may have during that session.

Each part will carry 50% of the full credit assigned for the final exam.

This structure for the final exam is to ensure that you have understood the methods that you have applied in solving these problems, and have not simply deployed an LLM to produce a solution for you without you putting any work in towards understanding that solution.

AI Use Permitted

To support your development and demonstration of the course learning outcomes, generative AI use in this course is permitted only as specified for each assignment. Each graded activity will identify one of four [AI Use Levels](#), from Level 0 through Level 3. Follow the stated level, including its permitted uses, prohibited uses, and documentation requirements. Basic grammar, spelling, and punctuation features, approved assistive technologies, and approved disability accommodations are permitted unless they generate or substantially rewrite content or replace the skill being assessed. Ask before using an AI tool if you are unsure whether it is permitted. Use of generative AI outside these boundaries may be considered a violation of Texas Tech University's academic integrity policy.

SCHEDULE (subject to change without notice)

Section	Topics	Reading	Notes
Software setup and quick introduction to git and GitHub	Installing Python, miniconda, and environment with necessary software for this course.	N/A	Week 1

Section	Topics	Reading	Notes
	• Quick introduction to git and GitHub. • Submitting homeworks using GitHub.		
Probability and Statistical Distributions	• Probability theory • Random variables • Probability and frequency • Central limit theorem • Generating random draws from arbitrary distributions	Ivezic, Ch. 1 & 3	Week 2—3
Frequentist Inference	• Point estimation • Least squares estimation • Maximum likelihood estimation • Bootstrapping and jack-knifing • Comparison of distributions	Ivezic, Ch. 4	Week 4—5
Bayesian Inference	• Priors • Parameter uncertainty quantification • Model selection • Conditional distributions • Marginalization	Ivezic, Ch. 5	Week 6—7
Data Exploration and visualization	• Non-parametric and parametric density estimation • Dimensionality reduction • Principal Component Analysis • Visualizing data	Ivezic, Ch. 6, 7	Week 8—9
Regression, parameter estimation, and model selection	• Formulating a model • Likelihoods • Markov chain Monte Carlo • Practical parameter estimation and model selection • Cross-validation techniques	Ivezic, Ch. 8	Week 10—13
Time-series analysis	• Deterministic and stochastic processes • Auto-correlation and cross-correlation • Structure function • Random Gaussian processes • Fourier power spectrum, Lomb-Scargle periodogram	Ivezic, Ch. 10	Week 14

Section	Topics	Reading	Notes
	Bayesian spectral estimation		
Deep Learning	- Neural networks - Fully connected, recurrent, and convolutional networks - Modern neural networks and LLMs. - Examples	Ivezic, Ch. 9	Week 15

ADA STATEMENT:

Any student who, because of a disability, may require special arrangements in order to meet the course requirements should contact the instructor as soon as possible to make any necessary arrangements. Students should present appropriate verification from Student Disability Services during the instructor's office hours. Please note: instructors are not allowed to provide classroom accommodations to a student until appropriate verification from Student Disability Services has been provided. For additional information, please contact Student Disability Services in Weeks Hall or call 806-742-2405.

ACADEMIC INTEGRITY STATEMENT:

Academic integrity is taking responsibility for one's own class and/or course work, being individually accountable, and demonstrating intellectual honesty and ethical behavior. Academic integrity is a personal choice to abide by the standards of intellectual honesty and responsibility. Because education is a shared effort to achieve learning through the exchange of ideas, students, faculty, and staff have the collective responsibility to build mutual trust and respect. Ethical behavior and independent thought are essential for the highest level of academic achievement, which then must be measured. Academic achievement includes scholarship, teaching, and learning, all of which are shared endeavors. Grades are a device used to quantify the successful accumulation of knowledge through learning. Adhering to the standards of academic integrity ensures grades are earned honestly. Academic integrity is the foundation upon which students, faculty, and staff build their educational and professional careers. [Texas Tech University ("University") Quality Enhancement Plan, Academic Integrity Task Force, 2010].

RELIGIOUS HOLY DAY STATEMENT:

"Religious holy day" means a holy day observed by a religion whose places of worship are exempt from property taxation under Texas Tax Code §11.20. A student who intends to observe a religious holy day should make that intention known in writing to the instructor prior to the absence. A student who is absent from classes for the observance of a religious holy day shall be allowed to take an examination or complete an assignment scheduled for that day within a reasonable time after the absence. A student who is excused under section 2 may not be penalized for the absence; however, the instructor may respond appropriately if the student fails to complete the assignment satisfactorily.

STATEMENT OF ACCOMMODATION FOR PREGNANT STUDENTS

To support the academic success of pregnant and parenting students and students with pregnancy related conditions, the University offers reasonable modifications based on the

student's particular needs. Any student who is pregnant or parenting a child up to age 18 or has conditions related to pregnancy may contact Alex Faris, the Texas Tech University designated Pregnancy and Parenting Liaison, to discuss support available through the University. The Liaison can be reached by emailing alfaris@ttu.edu. Should a student communicate with the instructor that they are pregnant or have a pregnancy related condition or may need additional resources related to pregnancy or parenting, the instructor will communicate that student's information to the Title IX Coordinator, who will work with the student and others, as needed, to ensure equal access to the University's education program or activity.

For more information regarding supportive measures, please contact pregnancy & parenting liaison Alex Faris (alfaris@ttu.edu | 806.834.3420) or visit [Pregnancy & Parenting](#). You can also [submit a Pregnancy & Parenting Support Form](#) to request assistance.

Use of Audio/Visual Recording Devices

NOTE TO INSTRUCTORS: Faculty may require a student who is recording the lecture as an accommodation to also sign an agreement that the recording will only be used for personal study use and will not be shared with other students or posted to any website or group.

Lectures, discussions, and instruction in a Texas Tech classroom may not be recorded via audio and/or visual means without prior permission from either the instructor or SDS, with prior notice to the instructor. Such permitted recordings are limited to legitimate academic purposes and cannot be shared with or distributed to any other person in any form without instructor approval. Students registered with Student Disability Services (SDS) may have an approved accommodation as documented on an SDS Letter of Accommodation that allows the student to record the class for their personal use to review course material.

Texas Tech faculty are entitled to copyright protection for their class materials and lectures, and unauthorized recordings may violate the faculty's copyright rights. Further, unauthorized audio and/or visual recording of academic classes may compromise the privacy rights of other students.

Violation of this policy may subject a student to discipline under the Texas Tech Code of Student Conduct.

Academic Misconduct

Texas Tech University expects students to "understand the principles of academic integrity and abide by them in all class and/or course work at the University" ([OP 34.12.5](#)). Academic misconduct includes but is not limited to cheating, plagiarism, collusion, falsifying academic records, misrepresenting facts, violations of published professional ethics/standards, and any act or attempted act designed to give unfair academic advantage to oneself or another student. Plagiarism is a form of academic misconduct that involves (1) the representation of words, ideas, illustrations, structure, computer code, other expression, or media of another as one's own and/or failing to properly cite direct, paraphrased, or summarized materials; or (2) self-plagiarism, which involves the submission of the same academic work more than once without the prior permission of the instructor and/or failure to correctly cite previous work written by the same student. [This video](#), retrieved from the University of Kansas Libraries website, provides an example of a plagiarism definition as well as examples of plagiarism and how to avoid it. Please review [Section B of the Code of Student Conduct](#) for more information related to other forms of Academic Misconduct, and contact your instructor if you have questions about plagiarism or other academic concerns in your courses. To learn more about the importance of Academic Integrity and practical tips for

avoiding plagiarism, explore the resources provided by the [TTU Library](#) and the [School of Law](#).

Civility In The Classroom

Texas Tech University is a community of faculty, students, and staff that enjoys an expectation of cooperation, professionalism, and civility during the conduct of all forms of university business, including the conduct of student–student and student–faculty interactions in and out of the classroom. Further, the classroom is a setting in which an exchange of ideas and creative thinking should be encouraged and where intellectual growth and development are fostered. Students who disrupt this classroom mission by rude, sarcastic, threatening, abusive or obscene language and/or behavior will be subject to appropriate sanctions according to university policy. Likewise, faculty members are expected to maintain the highest standards of professionalism in all interactions with all constituents of the university.

Discrimination, Harassment, and Sexual Violence

Texas Tech University is committed to providing and strengthening an educational, working, and living environment where students, faculty, staff, and visitors are free from sex discrimination of any kind. Sexual assault, discrimination, harassment, and other [Title IX](#) violations are not tolerated by the University. Discrimination or harassment based on race, color, or national origin (Title VI Misconduct) is also not tolerated by the University. Please report any incidents of Title IX or Title VI Misconduct to the Office for Student Civil Rights & Sexual Misconduct at (806) 742-SAFE (7233), titleix@ttu.edu, or file an [online report](#).

Faculty and staff members at TTU are committed to connecting you to resources on campus. Some of these available resources are:

- [TTU Student Counseling Center](#), 806-742-3674, (Confidential support on campus.)
- [TTU 24-hour Crisis Helpline](#), 806-742-5555, (Assists students who are experiencing a mental health or interpersonal violence crisis. If you call the helpline, you will speak with a mental health counselor.)
- [Voice of Hope Lubbock Rape Crisis Center](#), 806-763-7273, (24-hour hotline that provides support for survivors of sexual violence.)
- [Risk, Intervention & Safety Education \(RISE\) Office](#), 806-742-2110, (Provides a range of resources and support options focused on prevention education and student wellness.)
- [Texas Tech Police Department](#), 806-742- 3931, (To report criminal activity that occurs on or near Texas Tech campus.)

This list is not exhaustive, and the university offers a wide range of supportive measures in addition to those listed above. Please contact the Office for Student Civil Rights & Sexual Misconduct for more information.